Activity Series and Solubility Rules

Honors Chemistry reference sheet

Activity Series of the Metals

Most active at the top. Each metal is shown with the ion it forms.

Metal		Forms	Replaces hydrogen?
Lithium	**Li**	$Li^+ + e^-$	**React with cold water, steam and acids**
Cesium	**Cs**	$Cs^+ + e^-$	
Rubidium	**Rb**	$Rb^+ + e^-$	
Potassium	**K**	$K^+ + e^-$	
Barium	**Ba**	$Ba^{2+} + 2\ e^-$	
Strontium	**Sr**	$Sr^{2+} + 2\ e^-$	
Calcium	**Ca**	$Ca^{2+} + 2\ e^-$	
Sodium	**Na**	$Na^+ + e^-$	
Magnesium	**Mg**	$Mg^{2+} + 2\ e^-$	**React with steam and acids**
Aluminum	**Al**	$Al^{3+} + 3\ e^-$	
Manganese	**Mn**	$Mn^{2+} + 2\ e^-$	
Zinc	**Zn**	$Zn^{2+} + 2\ e^-$	
Chromium	**Cr**	$Cr^{3+} + 3\ e^-$	
Iron	**Fe**	$Fe^{2+} + 2\ e^-$	
Cadmium	**Cd**	$Cd^{2+} + 2\ e^-$	
Cobalt	**Co**	$Co^{2+} + 2\ e^-$	**React with acids**
Nickel	**Ni**	$Ni^{2+} + 2\ e^-$	
Tin	**Sn**	$Sn^{2+} + 2\ e^-$	
Lead	**Pb**	$Pb^{2+} + 2\ e^-$	
Hydrogen	**H_2**	**$2\ H^+ + 2\ e^-$**	***reference***
Copper	**Cu**	$Cu^{2+} + 2\ e^-$	**Do not replace hydrogen**
Silver	**Ag**	$Ag^+ + e^-$	
Mercury	**Hg**	$Hg^{2+} + 2\ e^-$	
Platinum	**Pt**	$Pt^{2+} + 2\ e^-$	
Gold	**Au**	$Au^{3+} + 3\ e^-$	

Halogens: $F_2 > Cl_2 > Br_2 > I_2$
most active first

Solubility Rules for Ionic Compounds in Water

Always soluble: compounds of the **Group 1** ions (Li^+, Na^+, K^+, Rb^+, Cs^+) and of ammonium, NH_4^+.

Compounds of these ions are soluble (aq) ...

Negative ion		... except with (insoluble)
nitrate	NO_3^-	*none*
acetate	$C_2H_3O_2^-$	*none*
chlorate	ClO_3^-	*none*
chloride	Cl^-	Ag^+, Hg_2^{2+}, Pb^{2+}
bromide	Br^-	Ag^+, Hg_2^{2+}, Pb^{2+}
iodide	I^-	Ag^+, Hg_2^{2+}, Pb^{2+}
sulfate	SO_4^{2-}	Ca^{2+}, Sr^{2+}, Ba^{2+}, Pb^{2+}, Hg_2^{2+}, Ag^+

Compounds of these ions are insoluble (s) ...

Negative ion		... except with (soluble)
carbonate	CO_3^{2-}	Group 1, NH_4^+
phosphate	PO_4^{3-}	Group 1, NH_4^+
hydroxide	OH^-	Group 1, Ba^{2+}
sulfide	S^{2-}	Group 1, NH_4^+, Ca^{2+}, Sr^{2+}, Ba^{2+}
sulfite	SO_3^{2-}	Group 1, NH_4^+
chromate	CrO_4^{2-}	Group 1, NH_4^+, Mg^{2+}, Ca^{2+}
silicate	SiO_3^{2-}	Group 1
oxide	O^{2-}	Group 1, Ca^{2+}, Sr^{2+}, Ba^{2+} *(these oxides react with water and form hydroxides)*

- Hg_2^{2+} is the mercury(I) ion.